

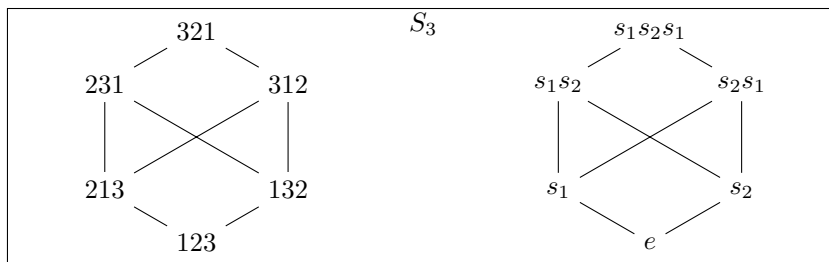
Bruhat order and applications: Lecture 3

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Today we will focus on the Bruhat order of a Coxeter group W . Bruhat order is especially important for applications of Coxeter groups to Lie theory (and especially Schubert calculus), which we will see a taste of today. Let's recall the definition.

Definition 0.1. Let u, v be elements of a Coxeter group W . Then $u \leq v$ in Bruhat order if there exists a reduced word $s_{i_1} \cdots s_{i_k}$ for v and a subword that is a reduced word for u .

To clarify the definition, a subword of $s_{i_1} \cdots s_{i_k}$ means the result of deleting some of the letters. For example, subwords of $s_1 s_2 s_3 s_2 s_1 s_2$ include s_3 , $s_2 s_1 s_2$, and $s_2 s_1$. (We cannot rearrange letters though: $s_2 s_1$ is not a subword of $s_1 s_2 s_3$.) Note that $u \leq_R v$ (comparing in weak order) implies that $u \leq v$ (comparing in Bruhat order). But there are usually more relations in Bruhat order. Here is the Bruhat order on S_3 .



Let's make two observations. The first is that $u \leq v$ holds if and only if $u^{-1} \leq v^{-1}$. This is a symmetry of Bruhat order that weak order doesn't have. The second is that, just like weak order, a cover relation $u < v$ in Bruhat order has the property that $\ell(v) = \ell(u) + 1$. Unlike weak order, this fact is nontrivial to prove. We'll do so later. First we will introduce a useful tool.

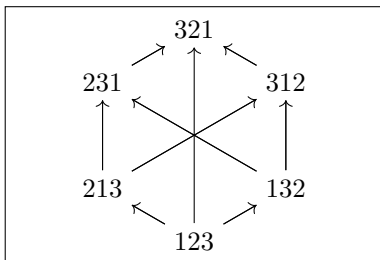
1 Bruhat graph

Definition 1.1. The **Bruhat graph** of a Coxeter group W is a directed graph with vertex set W and a directed edge

$$u \rightarrow v$$

if $\ell(u) < \ell(v)$ and there is a reflection $t \in T$ so that $u = tv$.

Here is the Bruhat graph for S_3 . Note that it is fairly easy to write down, since two permutations differ by multiplication by a reflection if and only if their one-line notations differ in exactly two spots.



To make use of the Bruhat graph, let's recall the properties that we proved on the last two worksheets.

Proposition 1.2. *Let W be a Coxeter group. Then the following properties hold.*

- (a) *Strong Exchange Property: If $t \in \text{inv}(w)$ and $s_{i_1} \cdots s_{i_k}$ is a reduced word for w , then there exists a j so that $tw = s_{i_1} \cdots \widehat{s_{i_j}} \cdots s_{i_k}$. (The hat notation means to omit s_{i_j} from the word.)*
- (b) *Exchange Property: If s is a descent of w and $s_{i_1} \cdots s_{i_k}$ is a reduced word for w , then there exists a j so that $ws = s_{i_1} \cdots \widehat{s_{i_j}} \cdots s_{i_k}$.*
- (c) *Deletion Property: If $s_{i_1} \cdots s_{i_k}$ is a non-reduced word for w , then there exist j and j' so that $w = s_{i_1} \cdots \widehat{s_{i_j}} \cdots \widehat{s_{i_{j'}}} \cdots s_{i_k}$.*

It turns out that the Bruhat graph tells us everything we need to know about Bruhat order.

Theorem 1.3. *We have $u < v$ if and only if there is a path from u to v in the Bruhat graph.*

To prove this, we will use a lemma.

Lemma 1.4. *If $u < v$, then there exists an element u' such that $u < u' \leq v$ with $\ell(u') = \ell(u) + 1$. Moreover, $u' = tu$ for some $t \in T$.*

Proof. Assume $u < v$. Let $s_{i_1} \cdots s_{i_k}$ be a reduced word for v . Then there is a subword which is a reduced word for u . Let j be the index of the first letter missing from the word; i.e. $s_{i_1} \cdots s_{i_{j-1}}$ are part of the word for u but s_{i_j} is not. There may be many subwords which are reduced words for u ; we pick one so that j is maximal. Let t be the reflection $s_{i_1} \cdots s_{i_j} \cdots s_{i_1}$. Then we get that $tu \leq v$, since t inserts a missing letter from v into the word for u . For the same reason, $\ell(v) \leq \ell(u) + 1$. We need to check that t is not an inversion of u ; in that case, $\ell(tu) = \ell(u) + 1$ and we have found our u' . Assume to the contrary that t is an inversion of u . Then t is in the standard inversion list for our reduced subword. If t is the j' 'th inversion for some $j' < j$, then $v = t(tv) = s_{i_1} \cdots s_{i_{j'}} \cdots s_{i_1}(tu)$. Since tu is the result of removing the letter s_{i_j} from v and $t(tu)$ is the result of further deleting the letter $s_{i_{j'}}$ from that, we have contradicted the reducedness of $s_{i_1} \cdots s_{i_k}$. So we must have that t is the j' 'th standard inversion for $j' > j$. But then since $u = t(tu)$ has a word which is the result of inserting s_{i_j} into the subword for u and then deleting a letter at position $j' > j$, we contradict the maximality of j . This is impossible, so we must have that t is not an inversion of u , so $\ell(tu) = \ell(u) + 1$. $\square$

Corollary 1.5. *If $u < v$ is a cover relation in Bruhat order, then $\ell(v) = \ell(u) + 1$.*

Proof of Theorem 1.3. ($\Leftarrow$) We just need to show that if there is an edge $x \rightarrow y$ in the Bruhat graph, then $x < y$. (This is because if there is a path $u \rightarrow x_1 \rightarrow \cdots \rightarrow x_r \rightarrow v$ then it follows that $u < x_1 < \cdots < x_r < v$). Assume $x = ty$ and $\ell(x) < \ell(y)$. Then t is an inversion of y . Let $s_{i_1} \cdots s_{i_k}$ be a reduced word for y . Then by the Strong Exchange Property, there is an index j so that $s_{i_1} \cdots \widehat{s_{i_j}} \cdots s_{i_k} = x$. Hence $x < y$.

($\Rightarrow$) Assume $u < v$. Then by applying Theorem 1.4, we find there is some u' with $u < u' \leq v$ and $\ell(u') = \ell(u) + 1$. Moreover, $u' = tu$, so there is a Bruhat graph edge $u \rightarrow u'$. If $u' = v$ then we are done. Otherwise, we can find a u'' with $u' < u'' \leq v$ and $u' \rightarrow u''$. By iterating this process, we find a path in the Bruhat graph from u to v . $\square$

Now let's notice that this result means we only need to check a single reduced word for v .

Corollary 1.6. *Pick a reduced word $s_{i_1} \cdots s_{i_k}$ for $v \in W$. Then an element $u \in W$ satisfies $u \leq v$ if and only if there is a reduced subword of $s_{i_1} \cdots s_{i_k}$ multiplying to u .*

Proof. One direction follows from the definition. To prove the other direction, let $s_{i_1} \cdots s_{i_k}$ be a reduced word for v . By Theorem 1.3, there is a path in the Bruhat graph from u to v . Let $t_1, \dots, t_k$ be the sequence of reflections used in the path. Then t_k is an inversion of v , so by the Strong Exchange Property $t_kv = s_{i_1} \cdots \widehat{s_{i_j}} \cdots s_{i_k}$ for some j . The resulting word might not be reduced, by the Deletion property we can delete further letters to make a reduced subword. Now, t_{k-1} is an inversion of t_kv , so we can repeat this process: delete one letter from the reduced subword, then use the Deletion Property to delete further letters. Continuing in this manner, we arrive at a reduced word for u which is a subword of $s_{i_1} \cdots s_{i_k}$. $\square$

This is sort of the opposite behavior from weak order; in weak order, to check $u \leq_R v$, you only need to pick one reduced word for u and check that it is a prefix of some reduced word for v . In Bruhat order, to check $u \leq v$, you only need to pick one reduced word for v and check that it has a subword multiplying to u .

2 Gale order

The Bruhat graph is great, but there are even better tools for checking Bruhat order in the symmetric group. To see it, we will first define a seemingly unrelated ordering on sets of integers.

Definition 2.1. Let X and Y be two subsets of $[n]$, both with size k . Then let the elements of X be written in order as $x_1 < x_2 < \dots < x_k$. Similarly, write the elements of Y in order as $y_1 < \dots < y_k$. Then we say that $X \leq Y$ in **Gale order** if $x_i \leq y_i$ for all $1 \leq i \leq k$.

For a permutation π , we write $\pi([k])$ to denote the set $\{\pi(1), \dots, \pi(k)\}$.

Theorem 2.2. Two permutations π_1 and π_2 in S_n satisfy $\pi_1 \leq \pi_2$ if and only if

$$\pi_1([k]) \leq \pi_2([k])$$

for all $1 \leq k < n$.

For example, $132 \leq 231$ since $\{1\} \leq \{2\}$ and $\{1, 3\} \leq \{2, 3\}$. But 4123 is not ≤ 1342 since $\{1, 3\} \not\leq \{1, 4\}$. This gives an easy way to check Bruhat order. In the problems, we will see an even easier way called the **Tableau criterion**.

3 Bruhat decomposition

In this section we will turn permutations into matrices by putting a 1 in column i row j if $\pi(i) = j$, and a 0 otherwise. This gives a representation of S_n called the standard representation.

Let $B \subseteq \text{GL}_n(\mathbb{C})$ be the subgroup of invertible upper-triangular matrices. The **Bruhat decomposition** says that

$$\text{GL}_n(\mathbb{C}) = \bigsqcup_{\pi \in S_n} B\pi B,$$

where we are writing π for the corresponding permutation matrix. In words, every invertible matrix can be written as

$$A = b_1 \pi b_2$$

for some upper-triangular invertible matrices $b_1, b_2 \in B$ and some permutation $\pi \in S_n$. Moreover, the permutation π is uniquely determined by A .

This is a more structured version of Gaussian elimination. Multiplying on the left by an upper-triangular matrix performs row operations in which row i can be rescaled and can have lower rows added to it. Multiplying on the right by an upper-triangular matrix performs column operations in which column j can be rescaled and can have earlier columns added to it. These operations are restrictive, but they are still strong enough to reduce every invertible matrix to exactly one permutation matrix.

How do we tell which permutation matrix a given matrix belongs to? The answer is to use ranks of certain submatrices. If A is an $n \times n$ matrix, define its **rank matrix** R_A by

$$R_A(i, j) = \text{rank (the submatrix of } A \text{ using rows } i, i+1, \dots, n \text{ and columns } 1, 2, \dots, j).$$

So $R_A(i, j)$ records the rank of the southwest corner of A whose upper-right entry is in position (i, j) .

For example, take the permutation $\pi = 231$. Its permutation matrix is

$$\pi = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

The rank matrix is

$$R_\pi = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 2 \\ 0 & 1 & 1 \end{pmatrix}.$$

For instance, the entry $R_\pi(2, 2) = 2$ because the submatrix using rows 2, 3 and columns 1, 2 is

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

which has rank 2. The entry $R_\pi(3, 1) = 0$ because the lower-left 1×1 submatrix is just (0).

Here is a less sparse matrix with the same rank matrix:

$$A = \begin{pmatrix} 2 & 8 & 11 \\ 1 & 7 & 23 \\ 0 & 1 & 6 \end{pmatrix}.$$

Indeed,

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 4 & 5 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \end{pmatrix},$$

so $A \in B\pi B$. Since multiplying on the left and right by upper-triangular invertible matrices does not change the ranks of these southwest submatrices, A and π must have the same rank matrix.

Conversely, the rank matrix determines the permutation. To see this for a permutation matrix, notice that

$$R_\pi(i, j) - R_\pi(i + 1, j) - R_\pi(i, j - 1) + R_\pi(i + 1, j - 1)$$

is equal to 1 exactly when the permutation matrix has a 1 in position (i, j) , and is equal to 0 otherwise. Here we interpret the missing entries outside the matrix as 0 in the evident way. Thus the rank matrix remembers where all the 1's of the permutation matrix are.

This explains the decomposition:

$$B\pi B = \{A \in \text{GL}_n(\mathbb{C}) \mid R_A = R_\pi\}.$$

The inclusion from left to right follows from the row and column operation interpretation above. For the other direction, one can run Gaussian elimination using only the allowed upper-triangular row and column operations; the ranks tell us where the pivots must occur, and those pivot positions form the permutation matrix π . Thus the matrices in $\text{GL}_n(\mathbb{C})$ break into pieces indexed by permutations, and the piece indexed by π consists exactly of the matrices with the same rank matrix as π .

If you want to continue learning about Coxeter groups, I recommend the books [1] and [2]. In particular, [1] has chapters specifically dedicated to weak order and Bruhat order. There are also many research papers on Coxeter group combinatorics that are readable; there is a tendency in this area to list every definition being used somewhere in the paper, even the definition of a Coxeter group, so they are often fairly self-contained.

4 Problems

1. Warm up: In S_4 , show that $s_1 s_2 s_1 \leq s_2 s_1 s_3 s_2$ and that the elements $s_1 s_2 s_3$ and $s_2 s_1 s_3 s_2$ are incomparable in Bruhat order.
2. Prove that there is a reflection $t \in T$ such that $u = tv$ if and only if there is a reflection $t' \in T$ such that $u = vt'$.
3. Prove, for $w \in W$, that the total number of incoming edges to and outgoing edges from w in the Bruhat graph is $|T|$.
4. How many total edges are there in the Bruhat graph of S_n ? How many total cover relations are in the weak order on S_n ?
5. Fix $1 \leq k < n$. A permutation is called **k -Grassmannian** if it is equal to e or has s_k as its unique descent.
 - (a) Show that taking the set $X_\pi = \{\pi(1), \dots, \pi(k)\}$ gives a bijection between k -Grassmannian permutations and subsets of $[n]$ of size k .

- (b) Show that k -Grassmannian permutations π_1, π_2 satisfy $\pi_1 \leq \pi_2$ if and only if $X_{\pi_1} \leq X_{\pi_2}$ in Gale order.
- (c) Find a bijection between subsets of $[n]$ of size k and partitions that fit in a box of size $k \times (n-k)$. Show that $X \leq X'$ in Gale order if and only if the corresponding partitions λ, λ' satisfy $\lambda_i \leq \lambda'_i$ for all $1 \leq i \leq k$.
- (d) Bonus: show that there is an analog of the Bruhat decomposition which says

$$\mathrm{GL}_n(\mathbb{C}) = \bigsqcup_{w \text{ is } k\text{-Grassmannian}} BwP_{n,k},$$

where $P_{n,k}$ is the group of matrices of the form $\begin{pmatrix} A & D \\ \mathbf{0} & C \end{pmatrix}$, where $A \in \mathrm{GL}_k(\mathbb{C})$ and $C \in \mathrm{GL}_{n-k}(\mathbb{C})$.

You saw in the Schubert calculus problem set that the Grassmannian $\mathrm{Gr}(k, n)$ is isomorphic to $\mathrm{GL}_n(\mathbb{C})/P_{n,k}$. Define the Schubert cell Ω'_λ to be $BwP_{n,k}/P_{n,k}$, where w is the k -Grassmannian permutation associated to λ . (Note this is not quite the same definition as in the Schubert calculus lectures; the Ω_λ defined there is in bijection with our Ω'_{λ^c} .) It follows from what you have shown in this problem (and the compactness of the flag variety) that $\Omega'_{\lambda_1} \subseteq \overline{\Omega'_{\lambda_2}}$ if and only if $\lambda_1 \leq \lambda_2$.

6. Show that the length generating function of S_n

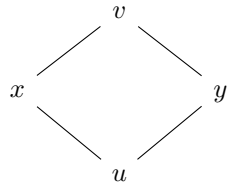
$$f(q) = \sum_{w \in S_n} q^{\ell(w)}$$

factors as

$$f(q) = (1+q)(1+q+q^2)(1+q+q^2+q^3) \cdots (1+q+\cdots+q^{n-1}).$$

The polynomial $1+q+\cdots+q^{k-1}$ is called a **q -integer** or **quantum integer**, denoted $[k]_q$. (This is because when you set $q=1$ you get the classical integer k .) The function $f(q)$ is called the **q -factorial**, denoted $[n]_q!$.

7. Prove that if $u \leq v$ and $\ell(v) = \ell(u) + 2$, then there are exactly two elements x such that $u < x < v$. Another way of phrasing this is to say that the **Bruhat interval** $[u, v]$ has the Hasse diagram

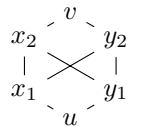


8. Prove that, in the symmetric group, $u \leq v$ if and only if $w_0 u w_0 \leq w_0 v w_0$ (this is in fact true for all finite Coxeter groups).
9. In the symmetric group, prove the **Tableau criterion**. This says that, to check that $u \leq v$, we just need to check that $\{u(1), \dots, u(k)\} \leq \{v(1), \dots, v(k)\}$ in Gale order for every k such that s_k is a descent of u .
10. In the symmetric group S_n , let's list all the transpositions as $t_1, t_2, \dots, t_{\binom{n}{2}}$ in lexicographic order, so that $t_1 = (1, 2), t_2 = (1, 3), \dots, t_{n-1} = (1, n), t_n = (2, 3), \dots, t_{\binom{n}{2}} = (n-1, n)$. Let $u < v$ be two permutations related by Bruhat order. Define a sequence of permutations by the following algorithm: set $u_0 = u$. At step i , do the following:

- If $u_{i-1} < t_i u_{i-1} \leq v$, then set $u_i = t_i u_{i-1}$.
- Otherwise, set $u_i = u_{i-1}$.

Prove that $u_{\binom{n}{2}} = v$. Moreover, show for all i that either $u_i = u_{i-1}$ or $\ell(u_i) = \ell(u_{i-1}) + 1$.

11. Assume $u \rightarrow v$ is an edge of the Bruhat graph such that $\ell(v) = \ell(u) + 3$. Prove that the Bruhat interval $[u, v] := \{x \in W \mid u \leq x \leq v\}$ has 6 elements, with Hasse diagram



Bonus: Prove the converse.

12. For each $w \in S_n$, show that there is an $\ell(w)$ -dimensional subgroup U_w of B such that the map $\phi : U_w \rightarrow \text{GL}_n(\mathbb{C})/B$ sending $u \in U_w$ to the coset uwB is bijective. (A consequence is that the Schubert cell BwB/B in the flag variety has complex dimension $\ell(w)$.)
13. The **Mobius function** of a poset P assigns to each pair $x \leq y$ an integer defined by

$$\mu(x, y) := \begin{cases} 1 & \text{if } x = y \\ -\sum_{x \leq z < y} \mu(x, z) & \text{if } x < y \end{cases}.$$

Show that Bruhat order on W is an **Eulerian poset**, meaning $\mu(u, v) = (-1)^{\ell(v) - \ell(u)}$ for all $u \leq v$.

References

- [1] A. Björner and F. Brenti, *Combinatorics of Coxeter Groups*, Graduate Texts in Mathematics, vol. 231, Springer, New York, 2005.
- [2] J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge Studies in Advanced Mathematics, vol. 29, Cambridge University Press, Cambridge, 1990.